

RUNZE ZHANG

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RESEARCH INTERESTS

My research leverages program analysis, graph theory, and machine learning to tackle large-scale cyber threats. My work spans a broad spectrum of challenges, including global botnet takedowns, mobile malware forensics, smart contract fraud detection, and click fraud mitigation.

EDUCATION

Ph.D. in Electrical and Computer Engineering

Jan. 2021 - Present

Georgia Institute of Technology

Atlanta, GA

[Cyber Forensics Innovation Laboratory](#)

Advisor: [Dr. Brendan Saltaformaggio](#)

M.S. of Cybersecurity

Sep. 2019 - Aug. 2021

Georgia Institute of Technology

Atlanta, GA

GPA: 3.88

B.E. in Electronic and Information Engineering

Sep. 2014 - May 2019

Hong Kong Polytechnic University

Hong Kong SAR, China

Thesis: *Real-time People Detection and Re-identification*

Advisor: [Dr. Kin-man Lam](#)

GPA: 3.83

PUBLICATIONS

Peer-Reviewed Top-Tier Security Conferences

Zhang, R., Sridhar, R., Yao, M. and et al., “Identifying Incoherent Search Sessions: Search Click Fraud Remediation Under Real-World Constraints,” to Appear in the *46th IEEE Symposium on Security and Privacy (S&P '25)*, San Francisco, CA, May. 2025. Acceptance Rate: 14.8%

Yang, Z., Chung, S., Chen, J., **Zhang, R.**, Saltaformaggio, B., Lee, W., ”COINDEF: A Comprehensive Code Injection Defense for the Electron Framework,” to Appear in the *46th IEEE Symposium on Security and Privacy (S&P '25)*, San Francisco, CA, May. 2025. Acceptance Rate: 14.8%

Zhang, R., Yao, M., Xu, H., Alrawi, O., Park, J., Saltaformaggio, B., “Hitchhiking Vaccine: Enhancing Botnet Remediation With Remote Code Deployment Reuse,” in *Proceedings of the 2025 Annual Network and Distributed System Security Symposium (NDSS '25)*, San Diego, CA, Feb. 2025. Acceptance Rate: 16.1% [[Open Source](#)]

NDSS Artifact Evaluation Result: 🌟Available, 🌟Functional.

Yao, M., **Zhang, R.**, Xu, H., Chou, R., Paturi, V., Sikder, A., Saltaformaggio, B., “Pulling Off The Mask: Forensic Analysis of the Deceptive Creator Wallets Behind Smart Contract Fraud,” in *Proceedings of the 45th IEEE Symposium on Security and Privacy (S&P '24)*, San Francisco, CA, May. 2024. Acceptance Rate: 17.8% [[Open Source](#)]

Media Coverage: [[Georgia Tech](#)]

Xu, H., Yao, M., **Zhang, R.**, Moustafa, M., Park, J., Saltaformaggio, B., “DVa: Extracting Victims and Abuse Vectors from Android Accessibility Malware,” in *Proceedings of the 33rd USENIX Security Symposium (Security '24)*, Philadelphia, PA, Aug. 2024. Acceptance Rate: 18.3% [[Open Source](#)]

USENIX Artifact Evaluation Result: 🌟Available, 🌟Functional.

Media Coverage: [[TechRadar](#)][[NY Breaking](#)][[MSN](#)] [[Hypepotamus](#)] [[hackerdose](#)] [[TechXplore](#)][[Sensi Tech Hub](#)] [[Georgia Tech](#)][[Science of Security](#)] [[WizCase](#)] [[Hackread](#)] [[xatakaen](#)]

INTERNSHIP EXPERIENCE

Research Sciences Intern *Microsoft Corporation*

May. 2023 – Aug. 2023
Redmond, WA

- Designed a deep learning model to detect search requests linked to click fraud for Bing.com.
- Prototyped session-level telemetry data from billions of daily searches and developed a *BERT-LSTM model* to detect click fraud from sessions' query-based semantic and temporal features.
- Analyzed JavaScript click-fraud malware binaries and integrated findings with *TF-IDF* and *entropy-based metrics* to build ground-truth datasets for model training and evaluation.
- Trained and benchmarked the prototype model with SOTA models, achieving 94% accuracy.
- Assessed the system's robustness to adaptive fraud strategies and documented the outcomes.
- The work is drafted, accepted, and to appear in *IEEE Symposium on Security and Privacy 2025*.

Application Developer

Application Technology Co. Ltd.

Jul. 2017 – Jun. 2018
Hong Kong SAR, China

- Developed a business card scanning IOS app, which combines *OCR* and *NLP* techniques to serialize contact information from card photos and sync with IOS built-in contacts.
- Implemented a *MongoDB-based* business card management system with RSA encryption.
- Built the *facial recognition* feature for a check-in IOS app and launched it to the Apple store.

RESEARCH EXPERIENCE

Graduate Research Assistant Georgia Institute of Technology

Jan. 2020 - Present
Atlanta, GA

Residential Proxy Detection via Network Activity Graphs | *Work In Progress* 2024

- Designed a *Graph Neural Network-based* model to identify residential proxy's IP for click fraud.
- Combined an IP probing framework with statistical analysis methods for ground truth datasets.
- Prototyped undirected graphs to link IPs with different types of network requests, including searches and clicks, and prepared the graph-based features for GNN models.
- Currently aim to perform real-time inference of residential proxy IPs for proactive click fraud detection on search engines. The prototype will be evaluated using Bing.com's real-world data.

Botnet Remediation via Remote Code Deployment Reuse | *Published - NDSS'25* 2024

- Developed a botnet remediation system that utilizes the frontend bot as input, extracting and repurposing its internal remote code deployment routines to run remediation code within bots.
- Developed a formal *graph-based modeling* with *hybrid program analysis* techniques to analyze payload deployment patterns and used *automated code generation* to build remediation code.
- Collaborated with *Netskope* on large-scale testing, successfully removing 523 real-world malware.

Creator Wallets Forensic Behind Smart Contract Fraud | *Published - S&P'24* 2023

- Built a forensic tool to trace fraud networks by analyzing behaviors of deceptive creator wallets.

- Developed a *graph-based model* integrating *symbolic analysis* to identify fraudulent contracts, achieving a 91% accuracy, uncovering 1M+ contracts responsible for 2.6M+ ETH illicit profits.
- Collaborated with *Etherscan*, showing the system's capability in proactively mitigating fraud.

Android Banking Accessibility Malware Analysis. | *Published - USENIX Sec' 24* 2023

- Systematically detecting and modeling the malicious behavior of Android malware abusing accessibility service to steal money from victims' banking apps for illicit profits.
- Utilized *symbolic execution* and *dyanmic force execution* approaches for malware sample forensic, revealing their targetted victims, infection vectors, and anti-analysis techniques.
- Deployed the pipeline on 9,850 malware samples, identifying 59K abuses and 215 targeted apps.

Undergraduate Research Assistant

Hong Kong Polytechnic University

Jun. 2018 - May 2019
Hong Kong SAR, China

Computer Vision and Image Processing-Related Research Projects.

2019

- Designed an image segmentation algorithm using the *U-Net CNN model* for detecting lung tumors in CT scans, achieving a *top-6 ranking* in the [2018 IEEE Video & Image Processing Cup](#).
- Built a HOG and CNN-based people detection and re-identification system as my honors project.

HONORS & AWARDS

The Internet Society 2025 NDSS Symposium Fellowship

2025

Awarded as one of the 31 [NDSS Symposium Fellows](#).

The Cisco Snort Scholarship

2021

Awarded \$10,000 for [The 2021 Snort Scholarship](#).

First Class Honours for Bachelor of Engineering

2019

Awarded First-Class Honours for outstanding Bachelor academic performance.

Hong Kong Polytechnic University Reaching-out Scholarship

2017

Awarded HK \$19,000 to support exchange-scholar education at McGill University.

TECHNICAL SKILL

Machine Learning: Transformers, Sequence Models, Graph Models, Tree-based Models.

Binary Reverse Engineering: Android Apps, JavaScript Web code, and Assembly Binaries.

Program Analysis: FlowDroid, Jadx, Ghidra, Angr, Frida, Wireshark, and IDA Pro.

Programming Languages: Python, Java, JavaScript, C#, C++, and SQL.

Development Tools and Frameworks: PyTorch, TensorFlow, pandas, MongoDB, Cosmos Bigdata platform, Docker, K8s, Kafka, Linux, Git, AWS.

INVITED TALKS

SkyWalker: Targets the Analysis of Command and Control Services of Botnets

TPCP Software Security Summer School, West Lafayette, IN

2020

PROFESSIONAL SERVICE

Artifact Evaluation Committee

The 34th USENIX Security Symposium

2025

Student Assistant

IEEE Secure Development Conference

2023

External Reviewer (Total = 21)

Network and Distributed System Security Symposium (NDSS)	2021 - 2025
IEEE Symposium on Security and Privacy (S&P)	2021 - 2025
USENIX Security Symposium(USENIX)	2021 - 2023
Annual Computer Security Applications Conference (ACSAC)	2020 - 2021
Digital Forensics Research Workshop (DFRWS)	2021 - 2023
ACM Conference on Computer and Communications Security (CCS)	2020
European Symposium on Research in Computer Security (ESORICS)	2020
International Symposium on Research in Attacks, Intrusions, and Defenses (RAID)	2020

REFERENCES

Professor Brendan Saltaformaggio
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Georgia Institute of Technology
Department of Electrical and Computer Engineering
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